Effects of Ambient Pressure and Nozzle Area Ratio on Thrust in a De Laval Nozzle

Abstract

This study investigated how ambient pressure and nozzle area ratio affect the predicted thrust of a de Laval nozzle. A one-dimensional, steady, ideal isentropic-flow model was implemented in Python using fixed chamber stagnation conditions and five nozzle area ratios (AR = 2–6). The supersonic exit Mach number was solved numerically, and the resulting exit pressure, temperature, velocity, mass flow rate, and thrust were calculated over 101 ambient-pressure values from 101.325 kPa to vacuum. The numerical solutions were cross-checked using Brent's root-finding method. For AR = 4, predicted thrust increased from 11.56 kN at sea-level ambient pressure to 15.61 kN in vacuum. Among the five geometries, AR = 2 produced the highest predicted sea-level thrust (12.55 kN), whereas AR = 6 produced the highest vacuum thrust (16.04 kN). Parametric CAD variants were built in Onshape to visualize changes in exit geometry. The work provides an idealized theoretical comparison rather than a validated prediction of a real nozzle.

1. Research Question, Hypothesis, and Objectives

Research question

How do ambient pressure and nozzle area ratio affect the thrust produced by a de Laval nozzle?

Hypothesis

It is hypothesized that the lower the ambient pressure, the greater the thrust produced by a de Laval nozzle will be. Furthermore, nozzles with smaller area ratios are expected to produce greater thrust at higher ambient pressures, while nozzles with larger area ratios are expected to perform better at lower ambient pressures.

Research objectives

- Investigate the influence of ambient pressure on predicted nozzle thrust using Python-based calculations.
- Compare the thrust of five nozzle area ratios under varying ambient-pressure conditions.
- Create three-dimensional, parametrically controlled nozzle geometries in Onshape.
- Evaluate the limitations of the ideal isentropic-flow model, particularly its omission of shocks, flow separation, and friction.

2. Methodology

2.1 Model assumptions and input parameters

Air was treated as a calorically perfect gas. The chamber values were prescribed for a comparative numerical example, not measured from a working engine. All five cases used identical throat area and chamber stagnation conditions.

Quantity	Symbol	Assumed value
Chamber stagnation pressure	P_0	1,000,000 Pa (1 MPa)
Chamber stagnation temperature	T_0	300 K
Throat area	A_t	0.0100 m²
Specific-heat ratio	γ	1.4
Specific gas constant	R	287 J/(kg·K)
Ambient pressure	Pamb	0–101,325 Pa, 101 points
Exit/throat area ratio	AR	2, 3, 4, 5, 6

The model assumes a choked throat (Mach 1), the supersonic area–Mach branch at the exit, steady flow, no heat transfer, no friction, and an isentropic path to the nozzle exit. These conditions are not asserted to hold for every real back-pressure condition [1–3].

2.2 Governing equations

The supersonic exit Mach number was determined from the area–Mach relationship [1]:

$$AR = \frac{A_e}{A_t} = \frac{1}{M_e}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M_e^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}}$$

Exit pressure, temperature, and velocity followed standard isentropic and perfect-gas relations [1]:

$$P_e = P_0\left(1+\frac{\gamma-1}{2}M_e^2\right)^{-\frac{\gamma}{\gamma-1}}$$

$$T_e = \frac{T_0}{1+\frac{\gamma-1}{2}M_e^2}, \qquad V_e = M_e\sqrt{\gamma R T_e}$$

The mass flow rate through a choked throat and the total nozzle thrust were computed as [1, 2]:

$$\dot{m} = A_t P_0 \sqrt{\frac{\gamma}{RT_0}}\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

$$F = \dot{m}V_e + (P_e - P_{amb})A_e$$

Symbols: M_e is the exit Mach number; A_e and A_t are exit and throat areas; P_e is exit static pressure; Pamb is ambient pressure; $\dot{m}$ is mass flow rate; and V_e is exit velocity. The momentum term ($\dot{m}V_e$) and pressure term ((P_e − Pamb)A_e) were kept separate in the calculations.

2.3 Numerical procedure and verification

Python was used to solve the area–Mach relation numerically. An initial grid of 10,000 candidate Mach-number values approximated the supersonic solution for each AR. These results were cross-checked using scipy.optimize.brentq, a bracketed root-finding method [4]. For all five cases, the differences between the grid approximation and Brent's solution were below 0.0004 in Mach number. Agreement between solvers checks numerical consistency for this equation; it does not validate the real-flow assumptions.

NumPy generated 101 uniformly spaced ambient-pressure values and supported numerical processing. Matplotlib was used to visualize the thrust curves. For each area ratio, $\dot{m}$, V_e, P_e and A_e were constant while ambient pressure was varied.

2.4 Parametric CAD modelling

Onshape was used to create a hollow, axisymmetric conceptual nozzle model. A two-dimensional converging–diverging wall profile was revolved through 360° around the longitudinal centerline. Five variants were saved at AR = 2, 3, 4, 5 and 6. The exit radius was linked to a dimensionless AR parameter:

$$r_e = r_t\sqrt{AR}$$

The throat area of 0.01 m² gives a radius of approximately 56.42 mm. A 320 mm total axial length (100 mm converging section, 220 mm diverging section), a 100 mm inlet radius, and an approximate 5 mm radial wall offset at reference locations were chosen as CAD assumptions. These lengths and the wall thickness were not optimized or structurally validated. CAD visualizes the area-ratio differences; it does not simulate fluid dynamics.

3. Results

3.1 Calculated flow properties

AR	Exit Mach	Exit pressure (kPa)	Exit temp. (K)	Exit velocity (m/s)
2	2.1972	93.93	152.63	544.12
3	2.6374	47.30	125.46	592.16
4	2.9402	29.79	109.93	617.94
5	3.1748	20.99	99.47	634.71
6	3.3679	15.84	91.78	646.76

The calculated choked mass flow rate was 23.34 kg/s for every geometry. As AR increased from 2 to 6, the predicted exit Mach number and velocity increased, while exit pressure and temperature decreased.

3.2 Predicted thrust and expansion pressure

AR	Exit radius (mm)	At 101.325 kPa (kN)	At vacuum (kN)	P_e = Pamb (kPa)
2	79.79	12.55	14.58	93.93
3	97.72	12.20	15.24	47.30
4	112.84	11.56	15.61	29.79
5	126.16	10.79	15.86	20.99
6	138.20	9.96	16.04	15.84

For AR = 4, predicted thrust increased from 11.56 kN to 15.61 kN as ambient pressure decreased from 101.325 kPa to 0 kPa. A negative linear relationship between thrust and ambient pressure was observed in this fixed-geometry ideal model.

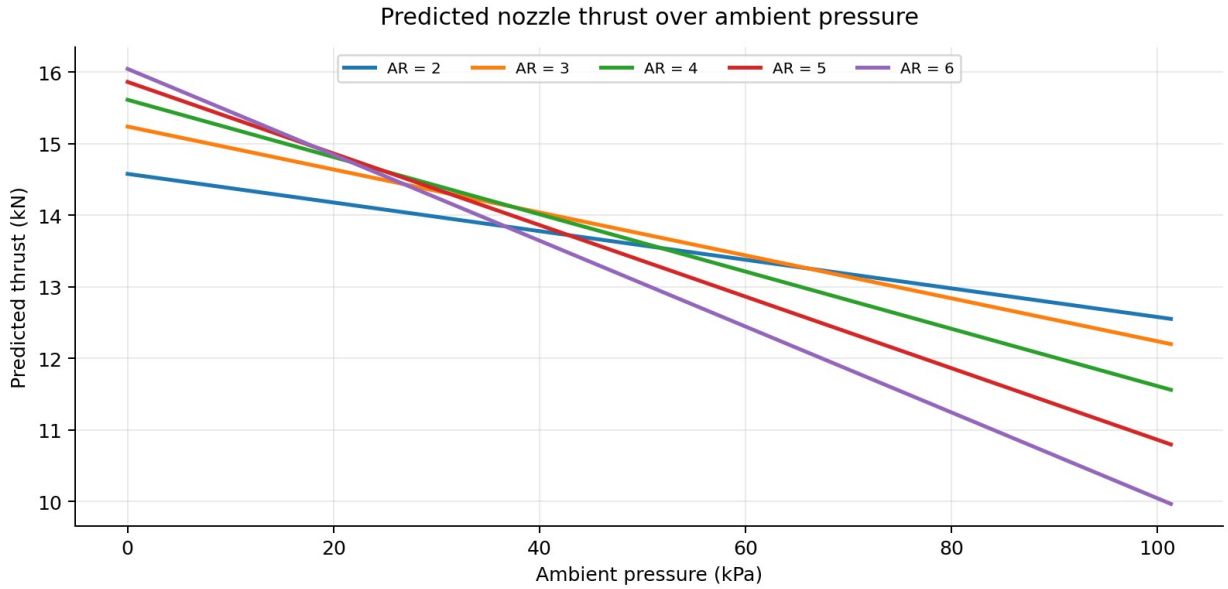


Figure 1. Predicted thrust versus ambient pressure for five nozzle area ratios. Computed from the ideal isentropic model; not measured data.

At 101.325 kPa, AR = 2 produced the highest predicted thrust among the five variants; in vacuum, AR = 6 produced the highest. No single area ratio in the investigated set maximized thrust over the entire pressure interval.

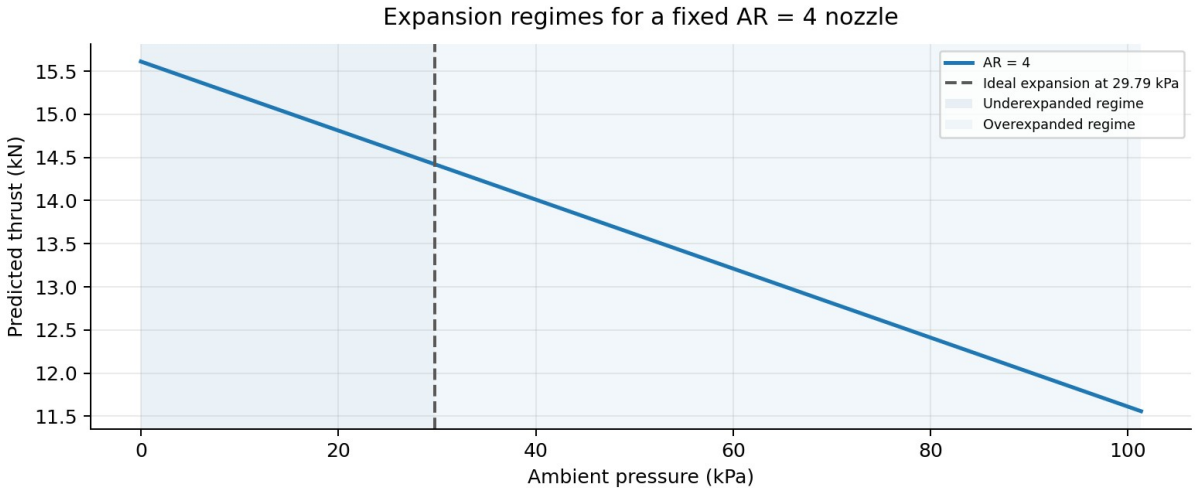
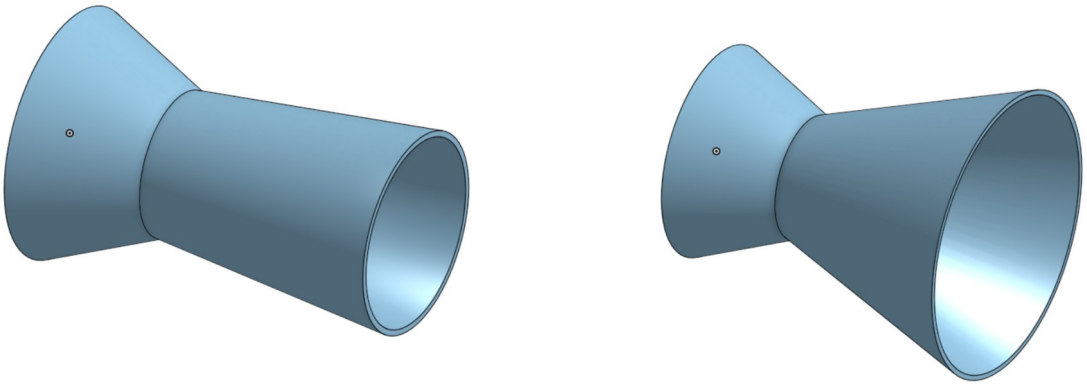


Figure 2. Idealized expansion regimes of the fixed AR = 4 geometry. The expansion boundary occurs when exit pressure equals ambient pressure (approximately 29.79 kPa).

3.3 CAD modelling outcomes

Five geometric states were stored by changing the AR parameter. The throat radius remained constant and the exit radius increased from 79.79 mm (AR = 2) to 138.20 mm (AR = 6). The images below illustrate the geometric change; they are CAD renders, not simulated pressure or velocity fields.



AR = 2 | exit diameter ≈ 159.58 mm *AR = 6 | exit diameter ≈ 276.40 mm*

Figure 3. Parametric CAD comparison using screenshots from the student's Onshape workspace. The two renders are illustrative, not dimensioned views.

A preliminary ISO A4 drawing was also created for AR = 2, including a longitudinal section (A–A). The nominal inner throat diameter is 112.84 mm and the exit inner diameter is 159.58 mm. An annotated drawing is included below as a supporting figure.

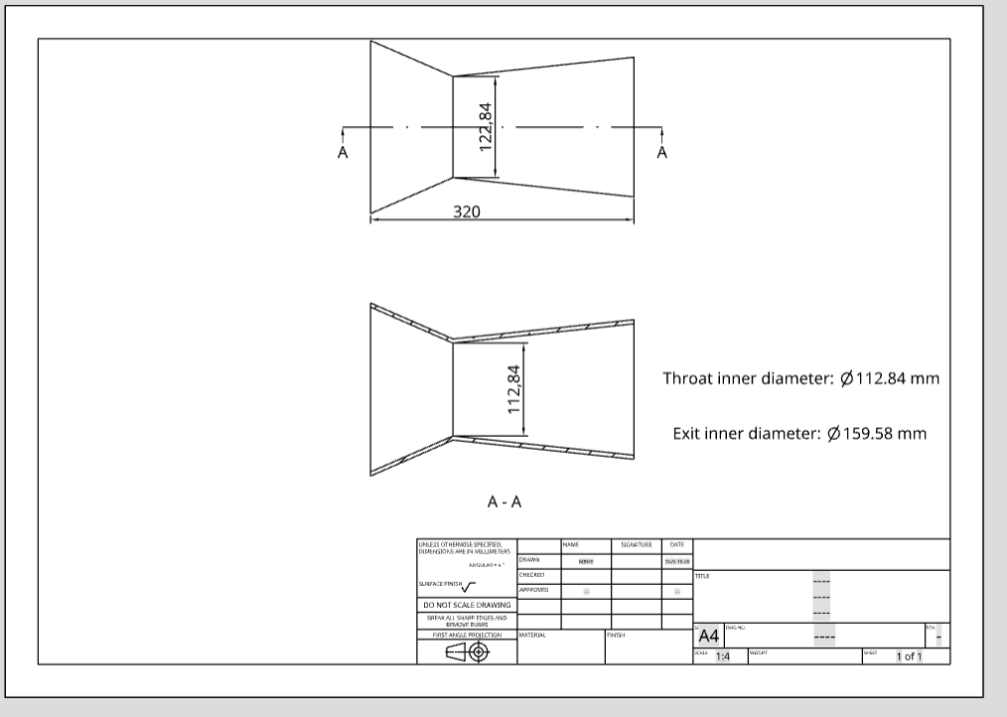


Figure 4. Preliminary AR = 2 engineering drawing (student-created Onshape screenshot). Manual annotations should be checked against the CAD model before final publication.

4. Discussion

4.1 Ambient pressure and area-ratio effects

The thrust increased as ambient pressure decreased because the difference between exit and ambient pressure became greater, raising the pressure-thrust contribution. For each fixed nozzle, momentum thrust remained constant in the ideal calculations because mass flow and exit velocity were fixed. Consequently, the predicted thrust–ambient-pressure curves were straight lines with slopes proportional to minus the exit area.

At sea-level ambient pressure, AR = 6 produced less thrust than AR = 2 even though its exit velocity was higher. The AR = 6 nozzle had a much lower exit pressure and a larger exit area, so the pressure-thrust contribution was strongly negative. In vacuum, ambient pressure was zero and AR = 6 obtained the higher total thrust largely because its higher exit velocity gave it greater momentum thrust, despite a smaller positive pressure-thrust contribution than AR = 2.

4.2 Limitations and interpretation

The model excludes shocks, flow separation, friction and boundary-layer effects; these may alter the pressure and velocity fields, especially in strongly overexpanded operation. The calculations also assume a fixed chamber stagnation state, constant specific heats for air, and ideal expansion to the exit over the full ambient-pressure interval. Actual nozzle behavior may depart from these assumptions [1–3].

The chosen 1 MPa chamber pressure, 300 K temperature, throat area, axial lengths and wall offsets were input assumptions rather than experimentally established design values. The CAD profile is conical and not an optimized contour; axial length and wall construction do not enter the present one-dimensional thrust equations. No CFD or physical testing was conducted. Therefore, the results support comparative ideal-model trends, not validated real-world performance or manufacturing readiness.

5. Conclusion

The results of this study support the hypothesis that, under ideal isentropic flow assumptions and fixed chamber conditions, predicted thrust increases as ambient pressure decreases. Furthermore, no single nozzle area ratio among the five investigated variants produces the maximum predicted thrust at every ambient pressure. The Onshape CAD models demonstrate that changing the nozzle area ratio modifies the exit radius while the throat radius remains constant. Future work could test the model against experimental or validated CFD data and investigate the sensitivity of the conclusions to the selected input assumptions.

References

[1] NASA Glenn Research Center. “Thrust Equations Summary.” https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/thrust-equations-summary/ (accessed 28 October 2026).

[2] NASA Glenn Research Center. “Mass Flow Rate Equations.” https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/mass-flow-rate-equations/ (accessed 28 October 2026).

[3] NASA Glenn Research Center. “Nozzle Design.” https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/nozzle-design/ (accessed 28 October 2026).

[4] SciPy. “scipy.optimize.brentq — SciPy Reference Documentation.” https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.brentq.html (accessed 28 October 2026).

Statement on AI assistance

AI assistance was used to explain and help develop portions of the Python workflow, review English-language phrasing, and organize the report. The student interpreted the outputs, worked through the physical relationships, and built the CAD variants using Onshape with guided assistance. This draft should be checked against the original notebook and revised to reflect the student’s final contributions and each university’s disclosure requirements.